MATH 2241 Assignment 1 Solution (Sept. 16, 2026)

1 (a) • Check (i)

There exists an integer $n \in \mathbb{Z}$ s.t.

$$n-1 < r \leq n \qquad \text{(Archimedean property)}$$

Then $\hat{r} \neq \mathbb{Q}$.

As $n-1 \in \hat{r}$, $\hat{r} \neq \emptyset$.

• Check condition (ii)

Let $x \in \hat{r}$ satisfying $y < x$.

$$x \in \hat{r} \Rightarrow x < r$$

By (O3), $y < r$ and so $y \in \hat{r}$.

• Check condition (iii)

Suppose m is the maximum in $\hat{r}$.

By denseness property of $\mathbb{Q}$, $\exists\, s \in \mathbb{Q}$ s.t. $m < s < r$.

Then $s \in \hat{r}$, contradicting m is the maximum.

(b) Let $\alpha = \{ x \in \mathbb{Q} : x^2 < 2 \text{ or } x \leq 0 \}$.

Details:

We now show $\alpha \neq \hat{r}$ for any $r \in \mathbb{Q}$.

It is shown in lecture note that

$$r^2 \neq 2 \quad \text{for any } r \in \mathbb{Q}.$$

Case 1: $r^2 < 2$.

It is shown in lecture note that

$$\exists\ q \in \mathbb{Q} \text{ s.t. } r^2 < q^2 < 2$$

This shows $\alpha \neq \hat{r}$

Case 2: $r^2 > 2$.

It is shown in lecture note that

$$\exists\ q \in \mathbb{Q} \text{ s.t. } r^2 > q^2 > 2.$$

Again $\alpha \neq \hat{r}$.

2. • Condition (i)

Let $a \in \alpha$, $b \in \beta$. Then $a+b \in \alpha+\beta$.

$\Rightarrow \alpha+\beta \neq \emptyset$.

Since $\alpha \neq \mathbb{Q}$ and $\beta \neq \mathbb{Q}$,

α and β are bounded above, say by M.

Then $\alpha+\beta$ is bounded above by $2M$

Then $\alpha+\beta \neq \mathbb{Q}$.

• Condition (ii)

Let $x \in \alpha+\beta$.

Suppose $x = a+b$ for some $a \in \alpha$ and $b \in \beta$.

Let $y < x$.

Let $n \in \mathbb{N}$ s.t. $\frac{1}{n} < x-y$.

Let $a' = a - \frac{1}{n}$, $b' = b - (x-y) + \frac{1}{n}$.

Then $a' \in \alpha$, $b' \in \beta$, and

$a'+b' = y$.

Thus $y \in \alpha+\beta$.

- Condition (iii)

Let $x \in \alpha + \beta$.

Then $x = a + b$ for some $a \in \alpha$ and $b \in \beta$.

By (iii), $\exists\ a' \in \alpha$ s.t. $a < a'$

$\exists\ b' \in \beta$ s.t. $b < b'$.

Then $a + b < a' + b'$ and $a' + b' \in \alpha + \beta$.

Thus x cannot be the maximum of $\alpha + \beta$.

Since x is arbitrary, $\alpha + \beta$ has no maximum.

3 (optional)

Condition (i)

Let $q < 0$. Then $q \in \alpha\beta \Rightarrow \alpha\beta \neq \emptyset$.

Since α, β are bounded above by $M > 0$

Then, $\alpha\beta$ is bounded above by M^2 and

so $\alpha\beta \neq \mathbb{Q}$.

Condition (ii)

Let $x \in \alpha\beta$.

Then $x = ab$ for some $a \in \alpha$ with $a > 0$

$b \in \alpha$ with $b > 0$.

Let $y < x$.

Let $n \in \mathbb{N}$ s.t. $x - y > \frac{b}{n}$.

Let $a' = a(1 - \frac{1}{n}) < a$

and $b' = \frac{y}{a(1-\frac{1}{n})} < \frac{x - \frac{b}{n}}{a(1-\frac{1}{n})} < b$.

Thus $a' \in \alpha$, $b' \in \beta$

$\Rightarrow y \in \alpha\beta$.

Condition (iii). Similar to addition part

4.. (Check u is Dedekind cut, optional)

Condition (i).

$\exists\ \alpha \in S \Rightarrow a \in \alpha \subseteq u \Rightarrow u \neq \emptyset$.

S is bounded above $\Rightarrow \exists\ \beta \in \mathbb{R}$ s.t.

$\alpha \subseteq \beta$, for all $\alpha \in S$.

Since β is bounded above by M for some $M \in \mathbb{Q}$,

u is also bounded above by M.

Condition (ii)

Let $x \in u$. Then $x \in \alpha$ for some $\alpha \in S$.

For $y < x$, by (ii) for α,

$y \in \alpha$.

$\Rightarrow y \in u$.

Condition (iii)

Let $x \in u$. Then $x \in \alpha$ for some $\alpha \in S$.

By (iii) for α, $\exists\ y \in \alpha \subseteq u$ s.t.

$x < y$

This implies x cannot be the maximum.

(Show u is the least upper bound of S.)

For any $\alpha \in S$, by definition

$$\alpha \subseteq u.$$

Thus, u is an upper bound of S

Let $\beta \subseteq u$ and $\beta \subsetneq u$.

Then $\exists\ x \in u$ and $x \notin \beta$, and so

$\exists\ x \in \alpha$ for some $\alpha \in S$, and $x \notin \beta$.

Thus $\alpha \not\subseteq \beta$, and so

β is not an upper bound of S.

In other words, u is the least upper bound of S.